INTERNATIONAL MONETARY FUND

KENYA

STAFF REPORT FOR THE 2014 ARTICLE IV CONSULTATION—DEBT SUSTAINABILITY ANALYSIS

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Kenya's debt outlook is favorable. Improvements in PV terms since the previous Debt Sustainability Analysis (DSA), published in 2013, reflect a higher discount rate. Kenya's risk of external debt distress remains low, while overall public sector debt dynamics continue to be sustainable.[1] *Under the baseline scenario and all the stress tests, Kenya's external debt burden indicators do not breach any of the relevant policy-dependent thresholds.*

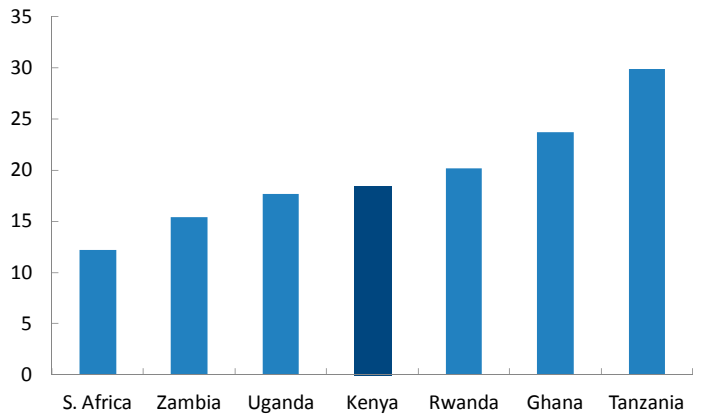
[1] The World Bank in 2013 upgraded its classification of Kenya to "strong performer" in terms of the quality of its policies and institutions as measured by a three-year average of the World Bank's Country Policy and Institutional Assessment (CPIA) Index. The relevant indicative thresholds for this category are: 50 percent for the NPV of debt-to-GDP ratio, 200 percent for the NPV of debt-to-exports ratio, 300 percent for the NPV of debt-to-revenue ratio, 25 percent for the debt service-to-exports ratio, and 22 percent for the debt service-to-revenue ratio. These thresholds are applicable to public and publicly guaranteed external debt.

BACKGROUND

1. **Kenya's overall public debt has been stable in recent years.** At end-2013, gross public debt totaled 41 percent of GDP, and has remained stable at around 41–42 percent of GDP since 2010. A little under half of Kenya's public debt is owed to external creditors. The DSA consists of two parts: external and public. The external DSA covers external debt of the central government and the central bank, as well as of the private sector; stress tests apply to public and publicly guaranteed by the government. The public DSA covers total debt—external and domestic—incurred or guaranteed by the central government. Public domestic debt comprises central government debt; government finance statistics are to be expanded to cover the newly created county governments. In this analysis, total public debt refers to the sum of public domestic and public external debt, but does not cover the entire public sector (e.g., parastatal borrowing without a government guarantee is not covered).

2. **Most of Kenya's external public debt remains on concessional terms, but its commercial component is increasing in 2014 with Kenya's first sovereign bond issuance.** Nominal public external debt as at end-2013 was USD 10 billion (text table), equivalent to 18 percent of GDP—broadly in line with comparable SSA countries (text chart). Multilateral creditors continued to account at end-2013 for more than 60 percent of external credit to Kenya. The share of debt from private creditors is set to increase in 2014, reflecting Kenya's successful inaugural

Figure 1: External Public Debt in Selected SSA Countries
(percent of GDP, 2013)



Sources: Country authorities, and IMF staff estimates.

Kenya: External Public Debt

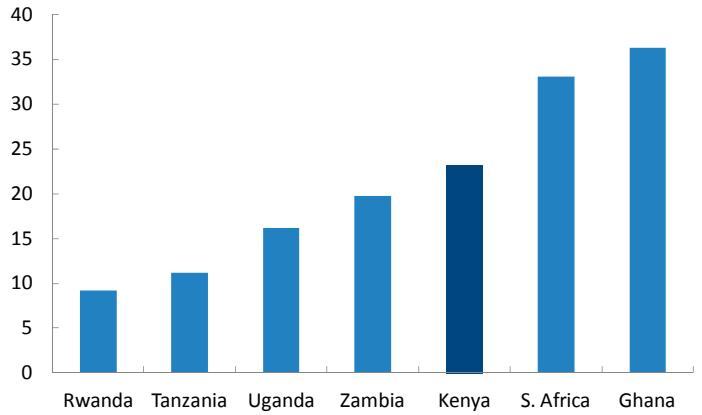
	2011		2012		2013	
	Billion USD	Share	Billion USD	Share	Billion USD	Share
Multilateral creditors	4.97	61.7	5.56	57.6	6.47	60.5
Bilateral creditors	2.29	28.4	2.71	28.1	2.84	26.6
Commercial banks	0.00	0.0	0.68	7.0	0.69	6.5
Others (supplier credits)	0.26	3.2	0.18	1.9	0.18	1.7
Total (excluding guarantees)	7.52	93.4	9.13	94.6	10.18	95.2
Publicly guaranteed debt	0.53	6.6	0.52	5.4	0.51	4.8
Total (including guarantees)	8.05	100.0	9.65	100.0	10.69	100.0

Source: Kenyan National Treasury.

sovereign bond issuance in June 2014. The USD 2 billion issuance comprises a 5-year bond totaling USD 500 million a 10-year bond totaling USD 1.5 billion. Kenya previously obtained commercial debt in 2012 through a two-year syndicated loan of about USD 600 million. In addition, in May 2014 Kenya signed an agreement with China featuring USD 3.8 billion in semi-concessional loans to finance construction of a regional railway. The authorities expect these loans to be disbursed over the next four years. Kenya did not seek debt relief under either the HIPC or MDRI initiatives.

3. **Kenya's gross domestic public debt[1] was around 23 percent of GDP at end-2013.** Domestic debt is issued mostly in the form of Treasury bonds (68 percent of domestic debt) and Treasury bills (27 percent). Commercial banks and nonbanks now hold roughly the same share of domestic debt (47 percent each); the central bank holds most of the remainder. The nonbank share has increased significantly over the past few years, reflecting greater diversification of the domestic investor base. Despite the relatively large size of the domestic debt (text chart), and some modest shortening of its maturities over the past two years, rollover risks remain moderate: the average maturity of Kenya's domestic debt was 5 years at end-2013, albeit down from 5.8 years in June 2011.

Figure 2: Domestic Public Debt in Selected SSA Countries
(percent of GDP, 2013)



Sources: Country authorities, and IMF staff estimates.

UNDERLYING ASSUMPTIONS

4. **This DSA is based on macroeconomic assumptions that are consistent with the framework outlined in the Staff Report for the 2014 Article IV consultation.** Notable revisions compared with the April 2013 DSA include (text table):

- Real GDP growth is weaker in short term but higher in the long term (Table 2), reflecting both the short-term impact of lower terms of trade but also sustained higher growth rates in the services sector.

  The primary fiscal deficit is projected to be around 2 percent of GDP wider in the short term owing largely to significant spending needs associated with devolution. In the medium term,

[1] Net domestic debt—gross domestic debt less government deposits at the central bank and at commercial banks, and government advances to parastatals—was slightly lower, at 21 percent of GDP at end-2013.

Kenya: Selected Macroeconomic Assumptions

	2012	2013	2014	2015	Long term
Real GDP Growth					
Current DSA 1/	4.5	4.6	5.0	5.8	7.0
Previous DSA (April 2013)	4.7	5.8	6.1	6.3	6.0
Primary Fiscal Deficit (percent of GDP)					
Current DSA	2.8	3.5	3.2	2.3	0.9
Previous DSA (April 2013)	2.2	2.0	1.6	1.4	1.1
Non-interest Current Account Deficit (percent of GDP)					
Current DSA	8.2	8.4	7.3	8.1	5.8
Previous DSA (April 2013)	8.7	7.1	7.8	7.1	3.9

Source: IMF staff estimates.
1/ Fiscal-year basis. The authorities expect to release upwardly revised GDP statistics in September 2014.

the authorities' commitment to regional integration implies a policy of deficit reduction.[1] In the long term, the primary deficit would be kept close to its previously projected level.

- The GDP deflator is projected to be higher in the medium term, reflecting pass-through from past falls in the terms of trade, though the terms of trade is projected to recover in subsequent years.

- Abstracting from the upward revision of GDP, the projected current account deficit is substantially wider reflecting larger historical deficits, current imports associated with oil exploration-related investment, and higher longer-term investment needs. The baseline does not include recent oil discoveries, pending confirmation that these will be commercially viable.

- The discount rate used to calculate the present value (PV) of external debt, now uses the unified rate of 5 percent, compared with 3 percent in the previous DSA. As flagged at the Sixth Review under the ECF (November 2013), the use of the revised (higher) discount rate results in a more favorable debt profile in PV terms.

- Indicative thresholds for debt are higher, owing to Kenya's improved CPIA rating.

[1] The EAC Monetary Union Protocol provides for fiscal convergence criteria including a ceiling on the fiscal deficit including grants of 3 percent of GDP; and a ceiling on the gross public debt of 50 percent of GDP in net present value terms. The other macroeconomic convergence criteria include a ceiling on headline inflation (8 percent) and a floor on reserve cover (4.5 months of imports). The fiscal plans outlined in the authorities' budget policy statement are consistent with Kenya meeting the EAC convergence criteria by 2021.

EXTERNAL DEBT SUSTAINABILITY ANALYSIS

5. **All external debt indicators remain well below the policy-dependent debt burden thresholds under the baseline scenario, and no thresholds are breached under any of the standard stress tests.** The main results of the external DSA are the following:

- **Under the baseline scenario, the debt burden remains sustainable over the 20-year projection period (Table 1a).** As a result of the issuance of the sovereign bond and railway-related loan package, the NPV of external debt would rise to 16 percent of GDP at end-2014 and stabilize at 16–17 percent (well below the 50 percent indicative threshold). The NPV of the debt-to-exports ratio would plateau at 90–95 percent in the medium term, remaining well under an indicative threshold of 200 percent.

- **Standard stress tests do not reveal significant vulnerabilities (Table 1b and Figure 1) as even the shocks with the highest impact would maintain debt levels below the relevant indicative thresholds.** The shock that would have the largest impact on external debt dynamics results from a one-time 30 percent nominal depreciation of the exchange rate (similar to the previous DSA) in 2014 and from a permanent shock to the terms of new public sector loans, increasing the PV of debt to GDP ratio from 17 to 26 percent in 2033, but still well below the relevant threshold.

- **The current account balance could improve significantly if recent resource discoveries are confirmed as profitable.** Kenya is currently a net oil importer. However, an oil company has reported estimated discovered resources of at least 600 million barrels of oil equivalent, and oil and gas exploration activities are continuing. If recent discoveries are confirmed as commercially viable, Kenya could become self-sufficient in 3–5 years and a net exporter in 5–10 years. This represents significant upside potential for the medium to long term.

PUBLIC DEBT SUSTAINABILITY ANAYLISIS

6. **Public debt peaks in 2015 and subsequently remains on a downward trend in the baseline scenario.** In 2014, overall public debt is projected to rise to 45 percent of GDP. The sovereign bond issuance and railway loan disbursements lift public debt to around 46 percent of GDP in 2015 (Table 2a). In subsequent years, on the basis of a primary deficit kept to 3 percent of GDP in 2016 and gradually brought down thereafter, public debt eases below 46 percent of GDP by 2018. In PV terms, the public debt-to-GDP ratio would remain stable around 41–42 percent through end-2018 (Table 2b), falling thereafter. The PV of public debt-to-revenue ratio gradually would decline from around 200 percent in 2014–15 to 191 percent in 2018.

7. **The alternative scenarios and bound tests indicate that the projected paths for all debt indicators remain under the relevant thresholds (Table 2b and Figure 2).** The shock with the highest negative impact on debt dynamics is real GDP growth one standard deviation below the historical average.

This shock pushes the ratio of PV of debt-to-GDP to 49 percent in 2018 compared with the baseline 41 percent; the PV of debt-to-revenue ratio to 229 percent from 191 percent, and the PV of debt service-to-revenue ratio to 30 percent from 26 percent by 2018. An unchanged primary balance from 2013 would have similar results. However, the authorities plan to reduce the primary balance in the medium term, consistent with the fiscal convergence deficit and debt criteria of the East African Community (EAC) monetary union.

8. **The devolution process has made concrete progress, but is also presenting challenges that—if left unaddressed—could have important medium-term fiscal implications.** Significant central government powers were devolved to 47 newly created county governments during the current fiscal year. Transitional challenges have emerged, as evidenced by capacity constraints at county level and problems in establishing adequate coordination between the two levels of government. From a general government perspective, underspending by counties could result in a lower budget deficit in the near term. However, staff cautions, and the authorities agree, the transitional challenges, together with devolution-related wage bill issues, the issue of the debt inherited from previous local authorities, and potentially imminent county borrowing, could—if left unaddressed—hamper efforts to reduce the primary balance in the medium term (even at the general government level).

9. **Contingent liabilities represent an additional fiscal risk.** Contingent liabilities in Kenya include publicly guaranteed debt of public entities, which reached KSh 43.5 billion (1 percent of GDP) at the end of 2012/13. About 15 percent of this guaranteed debt has been called. Public Private Partnerships (PPPs), mostly in the energy sector, represent an additional source of contingent liabilities that are yet to be fully assessed.

MAIN FINDINGS AND CONCLUSIONS

10. **This DSA finds that Kenya continues to face low risk of external debt distress.** Institutional improvements in recent years have increased the safety margin between Kenya's projected debt and the related indicative thresholds, though the use of a higher discount rate also contributes significantly to the more favorable dynamics. Upward revisions to GDP time series are also lowering the ratios of debt to GDP.

11. **Standard stress tests suggest scenarios in which debt would increase, but remain within sustainable bounds.** Exchange rate shocks and less favorable terms on new public sector loans represent the largest upside risks to external debt. Overall public debt would increase most in the event of significantly lower than anticipated economic growth. A scenario of this type could be triggered by the following adverse developments:

- a return to drought would reduce agricultural production and hydro-power generation, and would in turn reduce economic growth, increase food and oil-related imports and lead to a widening of the current account deficit.
- a deterioration in security conditions would weaken earnings from tourism, and dampen FDI inflows.

- a protracted slowdown in trading partner growth, in particular, in the euro area, coupled with declining commodity prices that would lower tourism and main export earnings (e.g. tea, horticulture and coffee), and reduce remittance and FDI inflows, with a significant impact on economic growth. However, in the external balances, these effects would be mitigated by an associated decline in oil prices.

- higher global food and fuel prices would raise the import bill and put pressure on the current account and the exchange rate.

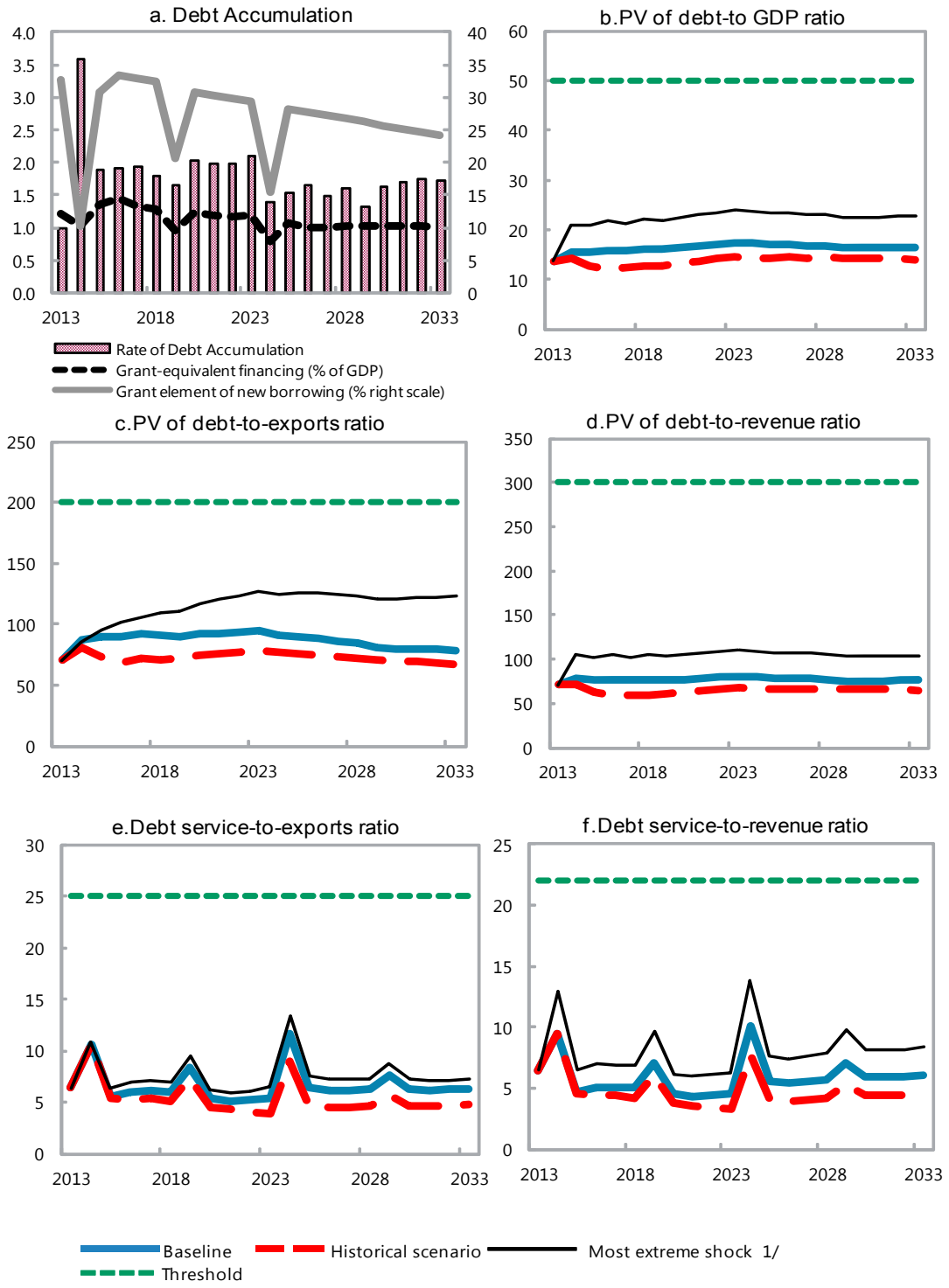
At the same time, Kenya has strong market foundations and long-standing sound macroeconomic policies—absence of price controls, flexible exchange rate and interest rates, limited budget subsidies—which give it scope to respond to shocks.

12. **The arrangements for the devolved institutions are only now being tested in practice.** The transitional problems of devolution could translate into fiscal risks for the central government if left unresolved, potentially affecting this year's budget outturns and creating pressures for the design of next year's budget. This makes it important for the authorities to maintain their efforts to reduce the primary deficit in the medium term, consistent with meeting the fiscal convergence criteria for the EAC monetary union.

13. **But natural resource discoveries represent upside potential for the medium to long term.** If confirmed, resource exports could significantly improve Kenya's external prospects. This puts a premium on appropriate design for policies to manage Kenya's natural resources.

14. **The authorities agree with the conclusions of this DSA.** They concur that Kenya is at low risk of external debt distress. They see Kenya's successful debut sovereign bond issuance in June 2014 as helping to diversify the country's funding sources and more broadly as a stepping stone in the path towards emerging market status. The authorities recognize specific fiscal risks in their medium-term budget policy statement, and have identified and are seeking to address the emerging challenges in the devolution process. The government has also initiated institutional and legal reforms for extractive industries.

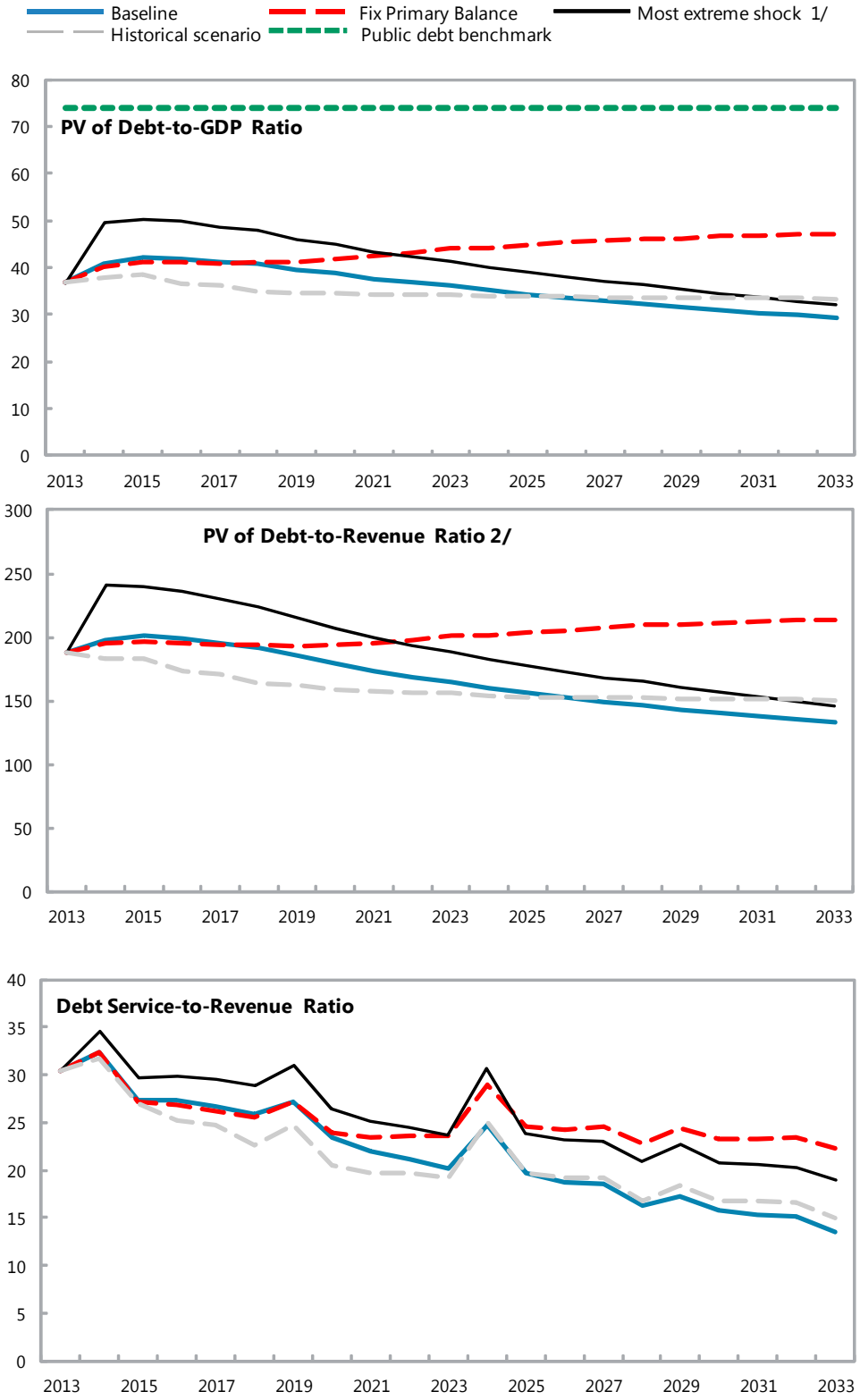
Figure 1. Kenya: Indicators of Public and Publicly Guaranteed External Debt under Alternative Scenarios, 2013-2033 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2023. In figure b. it corresponds to a One-time depreciation shock; in c. to a Terms shock; in d. to a One-time depreciation shock; in e. to an Exports shock and in figure f. to a One-time depreciation shock.

Figure 2. Kenya: Indicators of Public Debt Under Alternative Scenarios, 2013-2033 1/



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2023.
2/ Revenues are defined inclusive of grants.

Table 1a. Kenya: External Debt Sustainability Framework, Baseline Scenario, 2010-2033 1/
(In percent of GDP, unless otherwise indicated)

	Actual				Historical 6/ Average	Standard 6/ Deviation	Projections								
	2010	2011	2012	2013			2014	2015	2016	2017	2018	2014-2018 Average	2023	2033	2019-2033 Average
External debt (nominal) 1/	**26.9**	**29.3**	**30.7**	**30.7**			**33.0**	**33.9**	**35.3**	**35.7**	**37.1**		**41.8**	**43.1**	
of which: public and publicly guaranteed (PPG)	18.5	20.8	18.5	17.7			19.5	19.7	20.3	20.7	21.1		23.3	22.2	
Change in external debt	0.0	2.4	1.5	-0.1			2.3	0.9	1.4	0.4	1.4		1.1	0.3	
Identified net debt-creating flows	2.8	6.4	3.9	5.7			5.2	3.9	4.1	2.1	2.7		2.3	2.7	
Non-interest current account deficit	**5.2**	**8.7**	**8.2**	**8.4**	**4.6**	**3.2**	**7.3**	**8.1**	**6.7**	**6.2**	**5.6**		**5.8**	**5.9**	5.8
Deficit in balance of goods and services	11.3	15.0	13.8	13.7			13.2	14.0	13.0	12.4	11.8		11.8	11.1	
Exports	22.3	23.0	22.0	19.4			17.8	17.3	17.6	17.1	17.7		18.5	21.0	
Imports	33.6	38.0	35.8	33.1			31.0	31.4	30.6	29.5	29.5		30.3	32.0	
Net current transfers (negative = inflow)	-5.8	-6.2	-5.6	-5.7	-6.1	0.6	-6.3	-6.1	-6.0	-5.7	-5.8		-5.4	-4.8	-5.2
of which: official	-0.5	-0.4	-0.4	-0.4			-0.1	-0.1	0.0	0.0	0.0		0.0	0.0	
Other current account flows (negative = net inflow)	-0.3	-0.1	0.1	0.4			0.4	0.1	-0.2	-0.5	-0.4		-0.5	-0.3	
Net FDI (negative = inflow)	**-0.4**	**-0.8**	**-0.5**	**-0.9**	**-0.5**	**0.6**	**-1.8**	**-2.0**	**-1.9**	**-1.8**	**-1.9**		**-1.7**	**-1.0**	-1.5
Endogenous debt dynamics 2/	**-2.0**	**-1.5**	**-3.9**	**-1.8**			**-0.4**	**-2.2**	**-0.8**	**-2.3**	**-0.9**		**-1.9**	**-2.2**	
Contribution from nominal interest rate	0.2	0.2	0.3	0.3			0.4	0.4	0.5	0.5	0.5		0.6	0.6	
Contribution from real GDP growth	-2.1	-1.9	-0.4	-2.1			-0.8	-2.6	-1.3	-2.7	-1.5		-2.5	-2.8	
Contribution from price and exchange rate changes	-0.1	0.2	-3.7	-0.8			...	...	...	...	...		...	...	
Residual (3-4) 3/	**-2.8**	**-4.0**	**-2.4**	**-5.7**			**-2.9**	**-3.0**	**-2.7**	**-1.7**	**-1.3**		**-1.2**	**-2.4**	
of which: exceptional financing	0.1	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.3		-0.1	0.0	
PV of external debt 4/	...	...	26.5	26.6			29.0	29.8	30.8	30.9	32.0		35.9	37.5	
In percent of exports	...	...	120.5	137.0			163.1	171.8	174.6	180.8	180.9		193.9	178.5	
PV of PPG external debt	**...**	**...**	**14.2**	**13.6**			**15.6**	**15.6**	**15.8**	**15.8**	**16.0**		**17.5**	**16.5**	
In percent of exports	**...**	**...**	**64.7**	**70.1**			**87.4**	**89.9**	**89.7**	**92.8**	**90.7**		**94.3**	**78.7**	
In percent of government revenues	**...**	**...**	**75.6**	**71.4**			**77.5**	**76.0**	**76.4**	**76.3**	**76.3**		**80.9**	**75.9**	
Debt service-to-exports ratio (in percent)	**4.8**	**11.4**	**14.3**	**18.9**			**24.9**	**20.9**	**22.1**	**22.7**	**22.7**		**23.1**	**23.6**	
PPG debt service-to-exports ratio (in percent)	**3.9**	**3.7**	**3.9**	**6.4**			**10.6**	**5.6**	**5.9**	**6.1**	**5.9**		**5.3**	**6.3**	
PPG debt service-to-revenue ratio (in percent)	**4.5**	**4.6**	**4.6**	**6.5**			**9.4**	**4.7**	**5.1**	**5.0**	**5.0**		**4.6**	**6.1**	
Total gross financing need (Billions of U.S. dollars)	3.4	5.8	7.0	8.8			9.3	10.4	10.9	12.1	12.8		23.2	70.2	
Non-interest current account deficit that stabilizes debt ratio	5.2	6.3	6.7	8.5			5.0	7.2	5.3	5.8	4.2		4.7	5.7	
Key macroeconomic assumptions															
Real GDP growth (in percent)	8.6	7.6	1.7	7.5	5.1	3.0	2.8	8.8	4.1	8.9	4.4	5.8	6.6	7.2	7.0
GDP deflator in US dollar terms (change in percent)	0.4	-0.6	14.5	2.7	7.2	6.9	9.6	3.3	4.5	5.8	2.8	5.2	2.9	2.9	2.9
Effective interest rate (percent) 5/	0.9	1.0	1.0	0.9	1.3	0.5	1.3	1.5	1.5	1.5	1.5	1.5	1.6	1.5	1.5
Growth of exports of G&S (US dollar terms, in percent)	21.7	10.3	11.3	-2.7	12.0	11.1	3.3	9.4	10.6	11.6	11.2	9.2	11.2	11.2	11.3
Growth of imports of G&S (US dollar terms, in percent)	20.1	20.8	9.6	2.2	16.3	12.0	5.5	13.6	6.0	11.0	7.4	8.7	10.5	11.0	10.6
Grant element of new public sector borrowing (in percent)	...	...	...	32.7	...	...	10.3	30.8	33.4	32.9	32.4	28.0	29.3	24.2	26.2
Government revenues (excluding grants, in percent of GDP)	19.1	18.5	18.8	19.1			20.1	20.5	20.7	20.8	21.0		21.6	21.8	21.6
Aid flows (in Billions of US dollars) 7/	0.2	0.2	0.2	0.8			1.3	1.5	1.8	2.0	2.1		3.1	6.9	
of which: Grants	0.2	0.2	0.2	0.3			0.3	0.3	0.3	0.3	0.3		0.5	1.3	
of which: Concessional loans	0.0	0.0	0.0	0.6			1.0	1.2	1.5	1.7	1.8		2.6	5.6	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...	1.2			1.0	1.4	1.5	1.3	1.3		1.2	1.0	1.0
Grant-equivalent financing (in percent of external financing) 8/	...	...	...	44.7			18.6	40.5	40.7	39.5	38.9		36.1	31.8	33.7
Memorandum items:															
Nominal GDP (Billions of US dollars)	40.2	43.0	50.1	55.3			62.2	70.0	76.1	87.7	94.1		151.5	394.1	
Nominal dollar GDP growth	9.0	6.9	16.5	10.4			12.6	12.4	8.7	15.2	7.3	11.3	9.7	10.3	10.0
PV of PPG external debt (in Billions of US dollars)			7.0	7.5			9.5	10.7	12.0	13.5	15.1		26.4	65.0	
(PVt-PVt-1)/GDPt-1 (in percent)				1.0			3.6	1.9	1.9	1.9	1.8	2.2	2.1	1.7	1.7
Gross workers' remittances (Billions of US dollars)	1.0	1.4	1.9	2.3			2.5	2.8	3.0	3.4	3.7		5.6	12.9	
PV of PPG external debt (in percent of GDP + remittances)	...	...	13.7	13.1			15.0	15.0	15.2	15.3	15.4		16.8	16.0	
PV of PPG external debt (in percent of exports + remittances)	...	...	55.3	57.8			71.2	73.1	73.2	75.7	74.2		78.6	68.1	
Debt service of PPG external debt (in percent of exports + remittance	...	...	3.4	5.3			8.6	4.5	4.8	5.0	4.9		4.4	5.5	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt. For private sector debt, actual for 2010-11 and estimates for 2012-13.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 1b. Kenya: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2013-2033
(In percent)

	Actual	Projections						
	2013	2014	2015	2016	2017	2018	2023	2033
PV of debt-to GDP ratio								
Baseline	14	16	16	16	16	16	17	17
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	14	14	13	12	12	13	15	14
A2. New public sector loans on less favorable terms in 2013-2033 2	14	15	16	18	18	19	23	26
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	14	15	16	17	16	17	19	18
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	14	16	17	17	17	17	18	17
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	14	17	17	18	17	18	20	19
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	14	17	19	19	19	19	20	17
B5. Combination of B1-B4 using one-half standard deviation shocks	14	17	20	21	20	20	21	19
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	14	21	21	22	21	22	24	23
PV of debt-to-exports ratio								
Baseline	70	87	90	90	93	91	94	79
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	70	80	73	69	72	71	79	67
A2. New public sector loans on less favorable terms in 2013-2033 2	70	85	95	101	105	109	126	123
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	70	86	88	89	90	90	94	78
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	70	90	108	109	109	109	110	88
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	70	86	88	89	90	90	94	78
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	70	97	110	110	109	109	106	81
B5. Combination of B1-B4 using one-half standard deviation shocks	70	90	106	107	106	105	104	80
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	70	86	88	89	90	90	94	78
PV of debt-to-revenue ratio								
Baseline	71	78	76	76	76	76	81	76
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	71	71	62	58	60	60	68	65
A2. New public sector loans on less favorable terms in 2013-2033 2	71	76	80	86	87	92	108	119
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	71	76	80	82	79	82	86	81
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	71	78	82	84	81	83	85	77
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	71	83	84	86	83	86	91	85
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	71	86	93	94	90	91	91	78
B5. Combination of B1-B4 using one-half standard deviation shocks	71	87	99	100	96	97	98	85
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	71	105	102	105	102	105	111	104

Table 1b. Kenya: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2013-2033 (continued)
(In percent)

Debt service-to-exports ratio								
Baseline	6	11	6	6	6	6	5	6
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	6	11	5	5	5	5	4	5
A2. New public sector loans on less favorable terms in 2013-2033 2	6	11	5	6	6	7	8	9
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	6	11	6	6	6	6	5	6
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	6	11	6	7	7	7	7	7
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	6	11	6	6	6	6	5	6
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	6	11	6	7	7	7	7	7
B5. Combination of B1-B4 using one-half standard deviation shocks	6	10	6	7	7	6	6	7
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	6	11	6	6	6	6	5	6
Debt service-to-revenue ratio								
Baseline	6	9	5	5	5	5	5	6
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	6	9	5	4	4	4	3	5
A2. New public sector loans on less favorable terms in 2013-2033 2	6	9	4	5	5	6	7	9
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	6	9	5	5	5	5	5	7
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	6	9	5	5	5	5	5	6
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	6	10	5	6	6	6	5	7
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	6	9	5	6	6	6	6	7
B5. Combination of B1-B4 using one-half standard deviation shocks	6	10	5	6	6	6	6	7
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	6	13	6	7	7	7	6	8
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	22	22	22	22	22	22	22	22

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming
an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 2a. Kenya: Public Sector Debt Sustainability Framework, Baseline Scenario, 2010-2033
(In percent of GDP, unless otherwise indicated)

	Actual						Projections								
	2010	2011	2012	2013	Average 5/	Standard Deviation 5/	2014	2015	2016	2017	2018	2014-18 Average	2023	2033	2019-33 Average
Public sector debt 1/	40.8	42.0	41.0	40.8			44.7	46.2	46.5	46.1	45.9		42.0	34.9	
of which: foreign-currency denominated	18.5	20.8	18.5	17.7			19.5	19.7	20.3	20.7	21.1		23.3	22.2	
Change in public sector debt	1.2	1.2	-1.0	-0.2			3.8	1.5	0.3	-0.4	-0.2		-0.5	-0.5	
Identified debt-creating flows	1.3	-2.1	0.9	1.0			1.8	0.4	0.4	-0.6	-0.4		-1.1	-0.9	
Primary deficit	2.2	1.6	2.3	2.8	0.9	1.6	3.4	3.3	2.8	2.4	1.9	2.8	0.8	1.2	0.9
Revenue and grants	19.7	19.0	19.3	19.6			20.6	21.0	21.1	21.1	21.3		21.9	22.1	
of which: grants	0.6	0.4	0.4	0.5			0.5	0.5	0.4	0.3	0.3		0.3	0.3	
Primary (noninterest) expenditure	21.9	20.6	21.5	22.4			24.0	24.2	23.9	23.5	23.2		22.7	23.3	
Automatic debt dynamics	-0.9	-3.7	-1.4	-1.9			-1.6	-2.9	-2.4	-3.0	-2.3		-2.0	-2.1	
Contribution from interest rate/growth differential	-1.7	-3.1	-0.3	-1.4			-0.7	-2.6	-1.6	-2.8	-1.6		-1.7	-1.8	
of which: contribution from average real interest rate	1.4	-0.3	0.5	1.5			0.4	1.0	0.2	1.0	0.3		1.0	0.5	
of which: contribution from real GDP growth	-3.1	-2.9	-0.7	-2.9			-1.1	-3.6	-1.8	-3.8	-1.9		-2.6	-2.4	
Contribution from real exchange rate depreciation	0.8	-0.6	-1.1	-0.4			-1.0	-0.3	-0.8	-0.2	-0.7		...	...	
Other identified debt-creating flows	0.0	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	-0.1	3.3	-1.9	-1.1			2.0	1.1	-0.1	0.2	0.2		0.7	0.4	
Other Sustainability Indicators															
PV of public sector debt	...	...	36.8	36.8			40.7	42.1	42.0	41.3	40.8		36.1	29.2	
of which: foreign-currency denominated	...	...	14.2	13.6			15.6	15.6	15.8	15.8	16.0		17.5	16.5	
of which: external	...	...	14.2	13.6			15.6	15.6	15.8	15.8	16.0		17.5	16.5	
PV of contingent liabilities (not included in public sector debt)	...	...	...	...			...	...	...	...	...		...	...	
Gross financing need 2/	12.8	12.2	13.4	14.8			16.3	15.6	15.6	14.9	14.2		10.5	7.7	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	190.9	187.9			197.9	200.6	198.9	195.6	191.4		164.7	132.4	
PV of public sector debt-to-revenue ratio (in percent)	...	...	195.3	192.7			202.9	205.2	202.7	198.8	194.3		167.1	134.3	
of which: external 3/	...	...	75.6	71.4			77.5	76.0	76.4	76.3	76.3		80.9	75.9	
Debt service-to-revenue and grants ratio (in percent) 4/	25.5	26.2	28.0	30.5			32.4	27.4	27.3	26.6	25.9		20.2	13.5	
Debt service-to-revenue ratio (in percent) 4/	26.3	26.8	28.6	31.3			33.2	28.0	27.9	27.1	26.3		20.5	13.7	
Primary deficit that stabilizes the debt-to-GDP ratio	1.0	0.5	3.3	3.0			-0.4	1.8	2.5	2.8	2.1		1.3	1.7	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	8.6	7.6	1.7	7.5	5.1	3.0	2.8	8.8	4.1	8.9	4.4	5.8	6.6	7.2	7.0
Average nominal interest rate on forex debt (in percent)	1.0	1.2	1.3	1.3	1.5	0.5	2.0	2.3	2.4	2.4	2.4	2.3	2.8	2.9	2.8
Average real interest rate on domestic debt (in percent)	7.6	-0.7	2.6	7.1	1.6	3.9	1.7	4.1	0.3	3.7	0.9	2.1	3.8	2.1	3.7
Real exchange rate depreciation (in percent, + indicates depreciation	4.8	-3.5	-5.6	-2.6	-4.4	6.8	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	2.8	11.4	9.0	4.6	8.4	3.9	9.2	5.4	8.6	5.0	7.9	7.2	5.0	5.0	5.0
Growth of real primary spending (deflated by GDP deflator, in percer	0.1	0.0	0.1	0.1	0.1	0.0	0.1	0.1	0.0	0.1	0.0	0.1	0.1	0.1	0.1
Grant element of new external borrowing (in percent)	...	...	...	32.7	...	...	10.3	30.8	33.4	32.9	32.4	28.0	29.3	24.2	...

Sources: Country authorities; and staff estimates and projections.

1/ Refers to gross debt of the central government.

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 2b.Kenya: Sensitivity Analysis for Key Indicators of Public Debt 2013-2033

	Actual	Projections						
	2013	2014	2015	2016	2017	2018	2023	2033
PV of Debt-to-GDP Ratio								
Baseline	37	41	42	42	41	41	36	29
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	37	38	38	36	36	35	34	33
A2. Primary balance is unchanged from 2013	37	40	41	41	41	41	44	47
A3. Permanently lower GDP growth 1/	37	41	43	43	43	44	44	52
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2014-2015	37	41	47	48	48	49	49	48
B2. Primary balance is at historical average minus one standard deviations in 2014-2015	37	40	41	41	40	40	35	29
B3. Combination of B1-B2 using one half standard deviation shocks	37	39	41	42	42	42	41	39
B4. One-time 30 percent real depreciation in 2014	37	46	47	46	45	44	39	32
B5. 10 percent of GDP increase in other debt-creating flows in 2014	37	50	50	50	49	48	41	32
PV of Debt-to-Revenue Ratio 2/								
Baseline	188	198	201	199	196	191	165	132
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	188	183	183	173	171	163	156	150
A2. Primary balance is unchanged from 2013	188	195	196	195	194	194	201	213
A3. Permanently lower GDP growth 1/	188	200	205	206	206	205	201	237
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2014-2015	188	200	222	226	228	229	222	217
B2. Primary balance is at historical average minus one standard deviations in 2014-2015	188	194	194	193	190	186	161	130
B3. Combination of B1-B2 using one half standard deviation shocks	188	189	195	198	199	198	187	175
B4. One-time 30 percent real depreciation in 2014	188	224	223	219	213	207	176	146
B5. 10 percent of GDP increase in other debt-creating flows in 2014	188	241	240	236	230	224	188	146
Debt Service-to-Revenue Ratio 2/								
Baseline	30	32	27	27	27	26	20	14
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	30	32	27	25	25	23	19	15
A2. Primary balance is unchanged from 2013	30	32	27	27	26	26	24	22
A3. Permanently lower GDP growth 1/	30	33	28	28	28	27	24	24
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2014-2015	30	33	29	30	30	30	26	23
B2. Primary balance is at historical average minus one standard deviations in 2014-2015	30	32	27	27	26	25	20	13
B3. Combination of B1-B2 using one half standard deviation shocks	30	32	28	27	26	26	23	18
B4. One-time 30 percent real depreciation in 2014	30	35	30	30	29	29	24	19
B5. 10 percent of GDP increase in other debt-creating flows in 2014	30	32	30	33	31	31	23	15

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.